



Antibiotic Resistance Crisis: An Update on Antagonistic Interactions between Probiotics and Methicillin-Resistant *Staphylococcus aureus* (MRSA)

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Received: 10 August 2020 / Accepted: 1 March 2021

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Abstract

Antimicrobial resistance (AMR) havoc is a global multifaceted crisis endowing a significant challenge for the successful eradication of devastating pathogens. Methicillin-Resistant *Staphylococcus aureus* (MRSA) is an enduring superbug involved in causing devastating infections. Although MRSA is a frequent colonizer of human skin, wound, and anterior nares, the intestinal colonization of MRSA has greatly increased the risk of inducing MRSA-associated colitis besides creating a conducive environment for horizontal transfer of resistant genes to commensal microbes. On the other hand, staphylococcal resistance to last-resort antibiotics has urged the development of novel antimicrobial agents for the effective decolonization of MRSA. In response, probiotics and their metabolites (postbiotics) have been proposed as the adjunct therapeutic avenues. Probiotics exhibit a multitude of anti-MRSA actions (anti-bacterial, anti-biofilm, anti-virulence, anti-drug resistance, co-aggregation, and anti-quorum sensing) through the production of numerous antagonistic compounds such as organic acids, hydrogen peroxide, low molecular weight compounds, biosurfactants, bacteriocins, and bacteriocins like inhibitory substances. Besides, probiotics stabilize the epithelial barrier function and positively modulate the host immune system via regulating various signal transduction mechanisms. Preclinical and human intervention studies have suggested that probiotics outcompete with MRSA by exhibiting anti-colonization mechanisms via protective, competitive, and displacement mode. In this review, we aim to highlight the dynamics of MRSA associated virulence and drug resistance properties, and how probiotics antagonize MRSA through various mechanism of action.

Introduction

Antimicrobial resistance (AMR) is a global multifaceted menace faced by modern medicine for the past few decades. An alarming and persistent rise in antibiotic resistance among deadly pathogens is posing the greatest challenge by making their eradication a difficult task. The global emergence of AMR has been attributed to the irrational use of antibiotics in agriculture, human medications, and veterinary practices. Easy availability of antibiotics at low cost and few regulations governing antibiotic usage are some of the paramount factors driving antibiotics abuse [1]. The golden era of antibiotics was started with the discovery of penicillin

by Alexander Fleming in 1928. Since then, antibiotics have transformed modern medicine and saved millions of lives. However, shortly thereafter, penicillin resistance became a substantial clinical problem. In response, new beta-lactam antibiotics were discovered and developed [2]. However, further imprudent usage has boosted the bacterial adaptive phenomena to confront antibiotic selective pressure leading to the rapid emergence of multidrug-resistant superbugs like MRSA (Methicillin-Resistant *Staphylococcus aureus*) and Vancomycin-Resistant Enterococci (VRE), Extensively drug-resistant (XDR) *Mycobacterium tuberculosis*, Extended-Spectrum Beta-Lactamase (ESBL) possessing strains *Acinetobacter baumannii*, *Pseudomonas aeruginosa*, *Klebsiella pneumoniae*, and *Escherichia coli*, as well as *E. coli* and *Klebsiella pneumoniae* (KPC, *Klebsiella pneumoniae* carbapenemase) possessing New Delhi metallo- β lactamase 1 (NDM-1) [3]. The resistance among the superbugs is due to the intrinsic mechanisms such as the production of β -lactamase like enzymes, aminoglycoside-modifying enzymes to degrade antibiotics, efflux of antibiotics by

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alteration in outer membrane permeability, and modification in the antibiotics acting active sites. The acquisition of antibiotic-resistant genes (plasmids via horizontal gene transfer mechanisms) enables bacteria to survive under antibiotics selective pressure or by mutations that have evolved as part of a natural selection process that constitutes extrinsic or adaptive mechanisms. Importantly, virulome and resistome that are encoded in plasmid and mobile genetic elements (MGEs) are capable of transferring among commensal bacteria via horizontal gene transfer (HGT) [4]. Therefore, the development of new antibiotics will not provide an ultimate solution since the countdown to the emergence of resistance begins at the moment that the new antibiotic is introduced for clinical use [5]. Instead, unraveling the novel antimicrobials displaying multiple inhibitory actions are certainly warranted since multifaceted inhibitory actions indeed minimize the development of cross-resistance. In this context, probiotics, prebiotics, postbiotics, antibodies, bacteriocins, fecal microbial transplantation (FMT), bacteriophages, endolysins, and antimicrobial peptides (AMP) have been enlisted as alternatives to antibiotics (ATA) [6]. However, few of their limitations have restricted their applications. For example, the application of bacteriophages has been obstructed by their narrow-spectrum activity and unclear regulatory requirements. Moreover, the safety of phage therapy is a debatable issue as they show cytotoxicity and immunogenicity due to the presence of endotoxin and pyrogenic substances, depending on the phage preparation and route [6, 7]. Moreover, the phages that undergo only the lytic cycle are more desirable rather than the lysogenic cycle since the lysogenic cycle of phages may act as a vector to transfer potential antibiotic resistance genes (ARGs) from one host to another [8]. Antibody therapy is another alternative to tackle resistant infections [6]. Nevertheless, their high costs of production and poor shelf life have limited their application. Therefore, the present review focuses on highlighting the anti-MRSA potential of probiotics and their metabolites.

Methicillin-Resistant *Staphylococcus aureus* (MRSA)

Staphylococcus aureus is a Gram-positive, catalase-positive, coccus/round-shaped, aerobic bacterium arranged in the form of bunches and belongs to the family *Micrococcaceae*. According to the Center of Disease Dynamics, Economics and Policy (2019), approximately 36–42% of *Staphylococcus aureus* isolates found in India were resistant to oxacillin, 46–55% to amoxicillin-clavulanate, 94% to aminopenicillins, 53% to aminoglycosides, 4% to vancomycin, 90% to macrolides and 86% to carbapenems [9] which is largely due to its versatility and ability to acquire antibiotic resistance. MRSA is an extremely versatile human pathogen and

contributes significantly to skin microflora. Although burn wound and respiratory tract are commonly harbored by *S. aureus* [10], the gut mediated colonization of MRSA in the neonates and patients exposed to a prolonged hospital environment is also a significant threat posed by MRSA [11, 12]. Recently, the intestinal colonization tactics of MRSA with the typical manifestations of MRSA associated gastroenteritis and colitis (diffuse colitis and mild ileitis) have drawn attention globally [13]. Moreover, translocation of gut colonized MRSA to the wounds or infection site is highly worrisome [14]. MRSA continues to remain in the gut for months and evoke the epithelial barrier disruption and subsequently reach tissue and blood via bacterial translocation [15]. MRSA is known to cause varieties of infections ranging from mild skin infections to life-threatening diseases such as osteomyelitis, pneumonia, sepsis, endocarditis, bacteremia, toxic shock syndrome, and foodborne illness like food intoxication [16]. Such a notion of MRSA mediated disease burden was accounted for 72,444 invasive infections and 9,194 deaths in the United States alone during 2014 [17]. It is worth mentioning that the global pandemic of MRSA infections has troubled Americans more than that of combined acquired immune deficiency syndrome (AIDS), Parkinson's disease, emphysema, and homicide [18].

The genome of MRSA exhibits a distinct mobile genetic element (MGE), named staphylococcal chromosome cassette (SCC). SCC elements possess the *mec* genes encoding the resistance towards methicillin and other β -lactam antibiotics [19]. The presence of large SCC*mec* elements confers resistance to penicillin and their derivatives by modification of the serine site of transpeptidase of PBP2a, which is located in a deep pocket. However, they are highly diverse in their number (11 types), gene content, and in their size (~21 kb to 67 kb). SCC*mec* is integrated or located in the *orfX* gene (*orfX* gene ribosomal methyltransferase) without disrupting its expression [20]. The major feature of SCC*mec* is that it encompasses *mecA* gene complex along with its regulatory genes *mecI* and *mecR* in its cassette. The regions outside of the *ccr* and *mec* gene complexes are designated as “Joining” or “J” regions. SCC*mec* is flanked by specific invert repeats containing insertion sequences, which are recognized by *ccr*-encoded recombinases [21]. Further, the SCC*mec* also harbor phenol-soluble modulins (PSM), a kind of pro-inflammatory cytolytic toxin-producing toxin gene [22].

Infections of *S. aureus* are extremely versatile and spans depending upon the numerous virulence properties such as expression of a series of toxins, adhesion molecules, immune evasion, and other virulence determinants; which is also true with MRSA [23]. The genes associated with expressing the virulence determinants of *S. aureus* are categorized into two types viz. cell-surface-associated genes (adherence) and secreted (exotoxins) factors like toxins and hemolysins [24]. The exoprotein (protein A) coded by *spa* gene of *S. aureus*

facilitates the colonization to host epithelial cells [25, 26]. The coagulase (*Coa*) is another important virulence determinant of *S. aureus* that encodes the coagulase or clumping factors to promote bacterial growth inside the host and to invade host immune cells [27]. The whole-genome sequencing of *S. aureus* revealed the occurrence of genes related to enterotoxins, exotoxins, toxic shock syndrome (TSS) toxins, leukocidins, and leukotoxins production which is known to be involved in staphylococcal food poisoning and other functions [28]. *Nuc* is yet another virulent gene involved in encoding the thermostable nucleases that cleave the nucleic acids [29]. It has been reported that the production of the extracellular capsule encoded has been involved in playing a major role in causing infection by escaping the host immune system (phagocytosis) [30]. Besides, investigators have also reported several other virulent genes like *agrA*, *agrB*, *agrC*, *hla*, *saeR*, *odhB*, *lipA*, and *saeS* which are involved in the expression of hemolysin and other extracellular virulent enzymes [31]. Acquiring multiple drug resistance coupled with the expression of a series of virulence factors made *S. aureus* as historically a notorious superbug of greater public health significance.

Influence of Staphylococcal Biofilm Formation Trait on Antibiotic Resistance and Host Epithelial Colonization

Resistance towards multiple antibiotics is primarily a function of different adaptive phenomena such as alteration of drug target, impermeability to different antibiotics, and genetically associated changes (mutational events, genetic transfer of resistance genes through plasmids or mobile genetic elements, and mutations of target genes) [28]. Similarly, the formation of biofilms embedded in an exopolysaccharide matrix is one of the mechanisms by which bacteria acquire resistance to several antibiotics for survival under antibiotic selective pressure. In this regard, bacterial biofilms can be up to 1,000 times more resistant to antibiotics than that of a free planktonic state [32]. Biofilms are the ubiquitous sticky agglomerations of single or polymeric types of bacterial communities enclosed in self-produced polymeric matrices and adherent to inert or living surfaces. Almost all species of microorganisms can adhere to surfaces or among each other to form microbially diverse biofilms. Microbial biofilms pose significant clinical threats because the sessile bacterial cells gradually get inherently resistant to antimicrobial agents and host immunological activity due to resistance provided by exopolysaccharides. However, the extent to which organisms resist the action of different antimicrobials depends on the cell density, physiological state of the cells, and physical structure of the biofilm [33].

The biofilms produced by the *Staphylococcus aureus* are multilayered and embedded throughout the glycocalyx or slime layer with heterogeneous protein expression. This glycocalyx is primarily composed of 80% of teichoic acids and 20% of staphylococcal and host proteins [34]. *S. aureus* in the polymeric biofilms alters the metabolic and respiratory profiles like shifting from aerobic respiration to anaerobic respiration, expression of virulence and biofilm-related genes, etc. in the antibiotic selective pressures [35]. *S. aureus* produces the biofilm both on biotic and abiotic surfaces through three major steps viz. adhesion, proliferation, and detachment. During biofilm proliferation, it uptakes the nutrients by the channel in the biofilm which ultimately turns as a detachment factor for the detachment of biofilm from the surface and thereby helps in further spreading of biofilms. However, this detachment is strongly controlled by the quorum sensing phenomenon [36]. Harriott and Noverr [37] have reported that > 90% of the biofilms of *S. aureus* were resistant to vancomycin up to 1600 µg/mL compared to the free planktonic state. By contrast, several researchers have failed to observe the strong correlation between biofilm formation and antibiotic resistance among MRSA and MSSA isolates [38–40] and therefore further investigations are necessary to comprehend such mechanisms.

Gut: The Epicenter of Antibiotic Resistance

The gut is a residence of trillion commensal microflora which functions to ferment indigestible dietary fibers, production of various vitamins for epithelial cell nutrition, regulating epithelial cell proliferation and integrity, xenobiotic and drug metabolism, protect against various enteric pathogens, etc. [41, 42]. Microbial dysbiosis by antibiotic intervention leads to gut disorders with altered metabolic functions. Long-term administration of antibiotics might result in alteration of the gut microbiota by promoting the selection of a single bacterial community, most probably, the drug-resistant superbugs. This indeed increases the risk of horizontal gene transfer and finally turns the gut into a hub of multidrug-resistant genes [43]. On the other note, the gut colonization of superbugs like MRSA through the ingestion of contaminated food or other physiological activities might probably lead to the proliferation of only superbugs during antibiotic therapy. Moreover, the risk of horizontal gene transfer of potential ARGs from MDR pathogens to commensal microflora transforms the gut as an epicenter of antibiotic resistance [44]. In this regard, Piewngam et al. [45] have reported the MRSA colonization in the human gut since human feces manifested the presence of MRSA (10^3 – 10^4 cells). Such finding certainly underscores the failure of mainstay antibiotics in successful decolonization and thus demands the need for effective alternatives to antibiotics.

Probiotics as Living Drugs (Alternatives to Antibiotics)

The word ‘probiotic’ is derived from the Greek word ‘pro’ meaning ‘for’ and ‘bios’ meaning ‘life’ (Contrarily, the word antibiotic is also derived from the Greek word ‘anti’ meaning ‘against’ and ‘bios’ meaning ‘life’). Since time immemorial, the beneficial microorganisms are being frequently consumed in the form of fermented products for nutritional and therapeutic purposes without any scientific shreds of evidence [46]. However, such belief was conceptualized as probiotics by the Russian Nobel laureate Eli Metchnikoff. Probiotics may be defined as “Live microorganisms which when administered in adequate amounts confer a health benefit on the host” [47]. These include *Lactobacillus* (*L.*) *acidophilus*, *L. casei*, *L. lactis*, *L. helveticus*, *L. salivarius*, *L. plantarum*, *L. bulgaricus*, *L. rasmnosus*, *L. johnsonii*, *L. reuteri*, *L. fermentum*, *Bifidobacterium* (*B.*) *bifidum*, *B. breve*, *B. longum*, etc. and among yeasts, *Saccharomyces boulardii* is commonly used. Currently, other strains of bacteria like *Bacillus* spp., *Escherichia coli* strain Nissle 1917, certain enterococci, especially *Enterococcus faecium* SF68, *Wissella* spp. *Bacteroides xylanisolvans* DSM 23,694, *Bacteroides ovatus* D6, *Clostridium butyricum* MIYAIRI 588 have emerged as next-generation probiotics [48]. Hitherto, probiotics have been used as microbial biotherapies to manage inflammatory bowel disorder (IBD), pouchitis, ulcerative colitis, Crohn’s Disease (CD), lactose intolerance, etc. [49]. Besides, probiotics have shown enormous health benefits in a strain-dependent manner such as improvement of gut functions, alleviates lactose intolerance, enhances immune function, favorable reduction in risk of GI disease, gut barrier action, anti-carcinogenic effect (host epigenetic alteration), antidiabetic effect, anti-inflammatory effect, and so on [50, 51].

Mechanism of Action of Probiotics Against MRSA

Being biological response modifiers, probiotics assist in replenishing the gut microbial balance by effective displacement of pathogens and creating colonization resistance to pathogens. The possible antagonistic effect of probiotics is by adopting one or more of the following multiple mechanisms such as (i) Production of multiple metabolites have shown antagonistic effects on MDR pathogens [52] (ii) Biosynthesis of various biosurfactants and exopolysaccharides to overcome the microbial biofilms, the hallmark of antimicrobial resistance [53, 54]

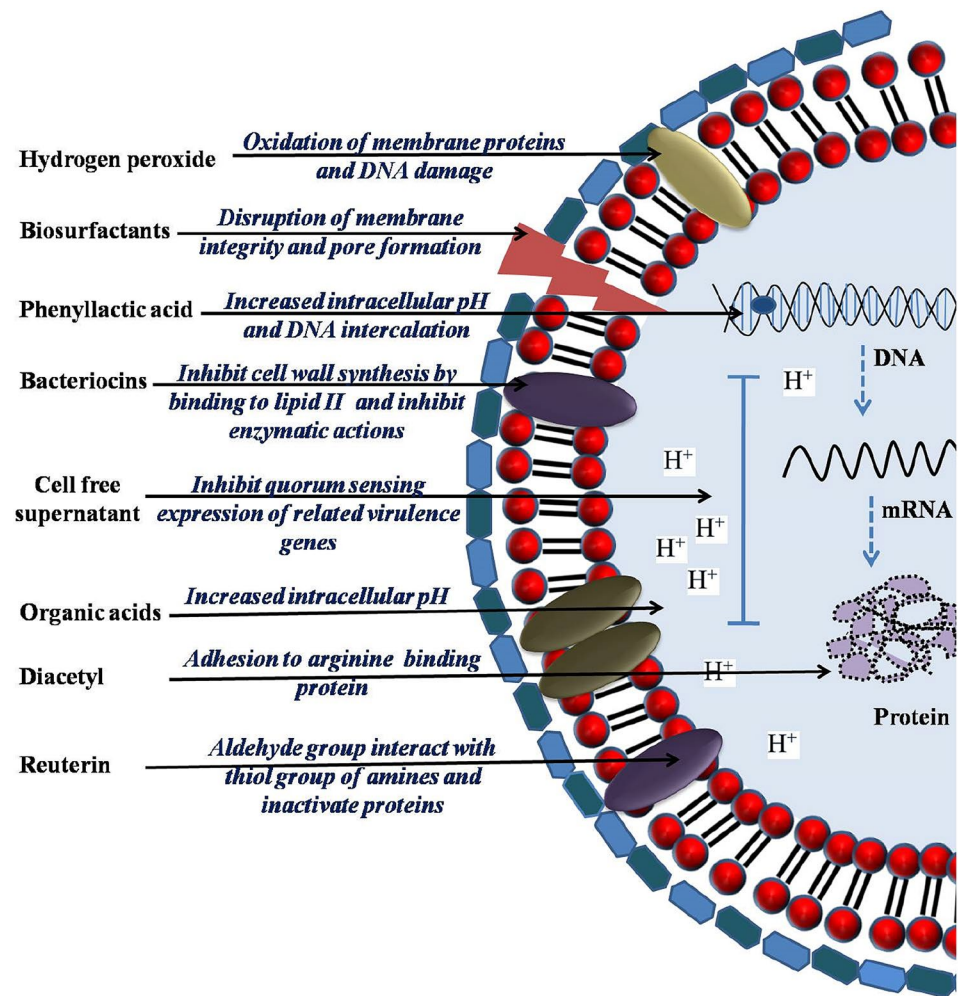
(iii) competition for nutrition and binding sites [55, 56] (iv) (vii) inhibition of quorum sensing by MDR pathogens, and thereby inhibit the virulence gene and other biofilm-related genes expression in pathogens. Boopathi et al. [57] reported that the probiotic strain viz. *Enterococcus durans* LAB38 exhibited the antibiofilm activity against MRSA and inhibition against Autoinducer 1 (A1) and Autoinducer 2 (A2) mediated quorum sensing phenomenon. Moreover, the intervention with *Bacillus* probiotics and *Bacillus* lipopeptides (fengycins) has evidenced the decolonization of *S. aureus* from the human gut by modulating Agr-based quorum sensing [45]. In another study, phenyllactic acid (PLA) derived from probiotic *Lactobacillus* strains has been shown to attenuate the virulence and pathogenicity of *Staphylococcus aureus* by interfering with the coordinated expression of different virulence factors [58]. However, such a quorum quenching mechanism of probiotics against AHL (N-acyl-homoserine lactones) of gram-negative drug-resistant pathogens has been well proved [59]. Such mechanisms of actions of probiotics against MRSA have been described in detail as below.

Antimicrobial Activity

The antagonistic knack of probiotics against multiple pathogens has made possible use probiotics as a preventive medicine to curb various diseases hitherto. As a result of cellular metabolism, lactic acid bacteria are capable of producing various antimicrobial compounds such as short-chain fatty acids and organic acids, hydrogen peroxide, ethanol, carbon dioxide and diacetyl, acetaldehyde, reuterin (low molecular weight compound), PLA, sorbic acid, etc. [60–62]. On the other hand, bacteriocins are the ribosomally synthesized antimicrobial peptides that inhibit the closely related organisms [63]. Indeed, bacteriocins and antimicrobials peptides have been classified separately as other alternatives to antibiotics to ameliorate the infections caused by antibiotic-resistant bacteria and to protect the commensal flora from the broad-spectrum antibiotics [64]. The various inhibitory mechanisms of probiotic bacterial derived bioactive compounds have been pictorially depicted in Fig. 1.

In this regard, Karska-Wysocki et al. [65] reported the antimicrobial action of probiotic strains viz. *Lactobacillus acidophilus* CL1285 and *Lactobacillus casei* LBC80R against MRSA #43 by displaying the inhibition zone around 2.93 ± 0.12 and 2.07 ± 0.12 cm, respectively. The purified biosurfactants of lactic acid bacteria have also shown the antimicrobial activity (around 14.6 mm diameter of inhibition) against various strains of MRSA viz. *S. aureus* (CdS, KR, CzR, PR, GR, AR, CS, NxR, NaR, MR, AkS, ES, TeS, TiS) [66]. Similarly, Gudina et al. [67] have reported that the biosurfactants extracted from *L. paracasei* ssp. *paracasei* A20 revealed effective inhibition of *S. aureus* at 25–50 mg/

Fig. 1 Diagrammatic representation on mode of inhibitory action of metabolites derived from probiotics on MRSA



mL. Bendajeddou et al. [68] investigated the antimicrobial activity of *L. paracasei* BMK2005 isolated from infant feces against MRSA and found remarkable antimicrobial activity against multidrug pathogens including MRSA isolates. The same investigators have also reported that the antimicrobial activity obtained was contributed by bacteriocin designated as paracaseicin A. Chen et al. [69] have observed that the combination of cell-free supernatant of probiotics strains viz. *L. salivarius* ATCC 11,741, *Bifidobacterium bifidum* ATCC 29,521, and *B. animalis* subsp. *lactis* BCRC 17,394 in combination with bovine lactoferrin exhibited greater inhibitory activity compared to the supernatant of probiotic strains alone against MRSA strains (1, 2, 4, 5, 19, and 46). *L. paracasei* ssp. *paracasei* M 27 isolated from the camel milk exhibited better antimicrobial property in the agar well diffusion method against MRSA [70]. Muralinath and Parasa [71] also showed the antibacterial activity *L. acidophilus* and *L. casei* against MRSA strains. The authors reported that *Lactobacillus acidophilus* and *Lactobacillus casei* demonstrated the highest inhibitory activity against all the tested strains of MRSA with zones of inhibition traverse from 20.0 ± 1.0 to

26.0 ± 1.25 mm (diameter). In another study, antimicrobial activity of probiotic strains like *L. plantarum* KF and *L. casei* Y1 was examined against MRSA which exhibited the inhibitory zone diameter of 23.3 ± 1.5 mm and 27 ± 1.0 mm, respectively [72]. These outcomes underscore the promising potentiality of probiotics derived cell-free supernatant (a consortium of derived inhibitory substances) to inhibit the growth of MRSA. Moreover, studying pharmacodynamic and pharmacokinetic properties of such probiotic derived biomolecules would help to develop better pharmacological drugs or skincare biotherapeutics to counterfeit the superficial infections of MRSA.

Antibiofilm Activity

Antibiofilm activity of probiotics is generally attributed to the extracellular production of a diverse class of biosurfactants that reduces the surface tension between interphase and prevents the bacterial adhesion (biofilm formation) or dislodges the preformed biofilms [73, 74]. On the other hand, the interaction of hydrophobic moieties of

biosurfactants with the bacterial membrane lipid affects membrane integrity by pore formation [75]. Indeed, such multifaceted antagonistic action of biosurfactants may effectively inhibit the superbugs that are resistant to conventional antibiotics with fewer probabilities of developing microbial cross-resistance. On the other hand, exopolysaccharides (EPS) produced by probiotic bacteria have also reported the knack to inhibit the adhesion of pathogenic microflora to gut econiche and by forming non-pathogenic biofilms in the gut. This strategy of self-biofilm formation by the probiotic bacteria with the aid of EPS enforced them to be recognized as “antibiofilm polysaccharides”. In this view, researchers reported the antibiofilm ability of EPS produced by *Bifidobacterium breve* BASO-1 and *L. fermentum* LB-69 against *S. aureus* ATCC 25923 [76].

Walencka et al. [77] have reported that the biosurfactants of three *L. acidophilus* strains exhibited ~ 50–60% reduction in biofilms by influencing staphylococcal cell surface hydrophobicity. In another study, Gudina et al. [67] reported $76.8 \pm 1.4\%$ reductions in biofilms of *S. aureus* by the biosurfactants of *L. paracasei* subsp. *paracasei* A20 at 50 mg/mL concentration. Kumar et al. [72] have investigated the antibiofilm ability of *L. casei* Y1 and *L. plantarum* KF isolated from *Tairu* (Dahi) and kefir against MRSA. The author reported that *L. casei* Y1 showing 3.54 log reduction in MRSA biofilm, whereas *L. plantarum* KF with 4.13 log reduction in MRSA biofilm. Importantly, biosurfactants prevent the colonization of pathogens to the food contact surfaces and human epithelial cells [78]. The in-depth evaluation further clued that probiotics may also affect at the molecular level by suppressing the expression of staphylococcal genes related to biofilm formation (*Ica* and *sarA*) (79 Zhang et al., 2021). Therefore, these findings support the exploitation of biosurfactants as bio-detergents in the biomedical field to get rid of staphylococcal biofilms on biological niches, surgical equipment, catheters, etc.

Anti-Quorum Sensing

Quorum sensing is a cell–cell communication process aided by signaling molecules that regulate the expression of several bacterial colonization and virulent genes after attaining a specific cell density. In *S. aureus*, the accessory gene regulator (*agr*) quorum-sensing system plays a crucial role in biofilm formation [79]. Nowadays, quorum sensing interference has been intensely studied as an anti-virulence strategy by exploiting various biomolecules to control MDR infections. The cell-free supernatant obtained from probiotic *B. subtilis* has suppressed the expression of the genes for staphylococcal quorum sensing (*argA*, *argB*, and *RNAIII*) [80]. Similarly, the supernatant of *Lactococcus lactis* has contributed to the down-regulation of *agr* related enterotoxins in *S. aureus* [81]. Cyclic dipeptides cyclo (L-Phe-L-Pro)

and cyclo(L-Tyr-L-Pro) molecules produced in the cultural supernatant of *Lactobacillus reuteri* RC-14 inhibited the staphylococcal quorum-sensing system (*agr*) and repressed the expression of TSST-1 in *S. aureus* MN8 [82]. Understandings from these studies suggested that endogenous microbiota like probiotics may be explored as biotherapeutics since they attenuate virulence factor production by *S. aureus*. However, further in vivo studies are warrant to support such findings.

Anti-Drug Resistance

Probiotics, despite exhibiting antibacterial activity, may also influence the expression and regulation of genes that confer antibiotic resistance to bacteria. Though limited attempts have been made to understand such mechanisms, an in vitro study underscores a significantly lowered expression of *mecA* (a gene that confers methicillin) following the treatment of MRSA with probiotic supernatant. In fact, the MRSA isolates treated probiotic supernatant after 70–120 h has shown susceptibility to the methicillin [83]. Such a mechanism of probiotics supports the synergistic use of cell-free supernatant and antibiotics that reduce the higher dose of antibiotic required to kill MRSA [84]. However, further studies at the proteome level are warranted to confirm the genome regulation on the expression and activity of PPB2a, the protein that confers resistance to penicillin (i.e., methicillin and oxacillin).

Antivirulence Ability

The co-treatment of MRSA with the cell-free culture supernatant of probiotic *Lactobacillus* spp, including *Lactobacillus reuteri*, *L. plantarum* and *L. fermentum* reduced the expression levels of *sea*, *sae*, *agr* A, *tst*, *spa*, and *spi* genes at different levels based on the concentration of cell-free supernatant used. In fact, 23.5 folds down-regulation of *sea* and *tst* genes was observed upon the incubation with 50% concentration of *L. reuteri* CFS after 12 h incubation [85]. Similarly, the treatment of *S. aureus* with the cell-free supernatant of probiotic *B. subtilis* suppressed the expression of adhesive molecules (*Cna* and *ClfA*) and virulence factor *Hla* [80]. In a co-culturing experiment, *Lactobacillus plantarum* inhibited the growth of *S. aureus* by ~ 1.5 log CFU/mL less than control and reduced the expression of staphylococcal enterotoxin (A, B, C, D, and E) levels by 76% [86].

Co-Aggregation

The co-aggregation behavior of probiotics plays important role in their interference with MRSA adhesion. It is a process of cell-to-cell aggregation either by the assistance

of cell surface appendages (pili, flagella, cell surface proteins) or due to electrostatic attraction or repulsion forces [87, 88]. Although the phenomenon of co-aggregation has been widely observed in all microbial systems, a genetically distinct co-aggregation knack of probiotic bacteria is merely advantageous since the collective evidences suggest that inter-bacterial flocculation by probiotic cells hampers the colonizing ability of pathogens or competitive cells. However, the extent of probiotic bacterial co-aggregation depends on cell surface hydrophobicity, cell surface molecules, co/aggregation-promoting factor, S-layer of proteins, etc. [89]. As can be seen in Table 1, several probiotic strains have shown the co-aggregation ability with *S. aureus* isolated from various sources. *Lactobacillus plantarum* ATCC 8014 and *L. brevis* gp104 showed 76–88.4% and 48% aggregation with *S. aureus* ATCC 25923, respectively [90]. This clearly indicates that two different probiotic strains can show different co-aggregation ability towards a single pathogen as the cell surface properties vary among the multiple probiotic strains.

Adhesion and Antiadhesion Knack of Probiotics Against MRSA

Bacterial adhesion to epithelial surfaces is a major criterion to select the putative probiotic strains possessing caliber to adhere to gut econiche at least transiently [91]. Probiotic adhesion to host epithelial cells through MAMP-PRR (microbe-associated molecular pattern-pattern recognition receptors) interaction results in host transcriptomic responses like immunomodulation. The cell-wall associated surface components such as teichoic acid, lipoteichoic acid, surface layer proteins, mannose-binding protein, fibronectin protein, mucous binding protein, etc. have been recognized as MAMP and pose a significant role in the adhesion of probiotic bacteria by recognizing PRR on the gut econiche [92]. The adhesion to epithelial cells is a complex process and is the function of physicochemical characteristics of the cell surface like bacterial cell surface hydrophobicity and electrical mobility/charge (rate of migration under electric field due to bacterial surface charges) [93]. Bacterial in vitro adhesion assays are used to ascertain their ability to colonize in the simulated gut system and also to overcome the possible difficulties in the in vivo animal model [94]. To investigate the in vitro adherence knack of probiotics, human epithelial cell models such as HT-29 and Caco-2 have been most commonly used since these cell lines express the same structural and functional features of normal human enterocytes [95]. Besides adhesion to gut enterocytes, their adhesion to mucin is a preliminary requirement of any probiotic strains to establish their colonization network in the gut system. In this context, investigators have

reported the strain-specific behavior of 18 different strains of *L. plantarum* in mucin adhesion, wherein average adhesion to mucin varied from ~50–70% [96]. In another study, investigators have reported the 30–40% adhesion of probiotic *Lactobacillus* strains viz. *L. paracasei* K5, *L. rhamnosus* GG, and *L. casei* ATCC 393 to epithelial cell line [97]. In a comparative study on adhesion of *Lactobacillus acidophilus* DDS-1 (human origin), *B. animalis* ssp. *lactis* UABla-12 (human origin), *L. plantarum* UALp-05 (plant origin), and *Streptococcus thermophilus* UASSt-09 (dairy origin) to HT-29 cells, the human origin probiotic strain viz. DDS-1 (87–93%) exhibited superior adhesion, which was followed by UALp-05 (89–92%) and UABla-12 (79–86%), which indicated the potency of autochthonous probiotic strains in adhesion to human colon cells [95].

Bacterial interference creates an ecological competition between probiotic bacteria and pathogenic species (Fig. 2). Despite the antibiofilm activity, probiotics have been found to exhibit anti-adhesion potentiality towards various pathogens by acquiring various mechanisms like displacement of adhered pathogens from the gut (exclusion) or creating competition with pathogens for the binding site (competition) or by self-adherence of probiotics to gut and thereby inhibiting the other pathogens to colonize (protective) [6]. The mechanistic insights by which probiotic bacteria exhibit the pathogen exclusion are by the creation of a hostile microecology, elimination of available bacterial receptor sites by self-adhesion, secretion of antimicrobial compounds, and selective metabolites, and competitive depletion of essential nutrients for competitive microflora [99]. It was also reported that some of the probiotic lactobacilli and bifidobacteria share common carbohydrate-binding specificities with some pathogens [96, 97] which makes it possible for them to compete with specific pathogens for the receptor sites in the host gut. In nutshell, probiotic strains can inhibit the attachment of pathogenic bacteria through steric hindrance at enterocyte pathogen receptors. It was also proved that probiotic strains with maximum cell surface hydrophobicity, auto-aggregation, and co-aggregation caliber possess the maximum tendency to eliminate *S. aureus* [98].

A consortium of probiotic strains viz. *L. rhamnosus* GG, *L. rhamnosus* LC705, *Bifidobacterium breve* 99, and *P. freudenreichii* JS has displayed 28.3% inhibition in the adhesion of *S. aureus* to human gastric mucin, 45% by competitive assay and 30.1% by protective mode [99]. Ren et al. [100] investigated the anti-colonization efficacy of probiotic strains in Caco-2 cell line and reported 44% and 74% reduction in the adhesion of *S. aureus* by *L. salivarius* and *L. plantarum* in the protective assay, 30% and 70% prevention of adhesion *S. aureus* in the competitive assay and 2% and 30% of displacement of the adhered *S. aureus* to Caco-2 cell line. The same investigators reported that the surface layer proteins (Slps) of *L. salivarius* exhibited

Table 1 In vitro and In vivo studies indicating antagonistic actions of different probiotic strains against MRSA

Probiotic strain	Assay	Component	Outcome	References
<i>Lactobacillus acidophilus</i> , <i>Lactobacillus bulgaricus</i> , <i>Lactobacillus casei</i> , <i>Bifidobacterium bifidum</i> , <i>Bifidobacterium animalis</i> , <i>Lactobacillus plantarum</i> , <i>Streptococcus thermophilus</i>	Agar well diffusion assay	Cell free supernatant	Antimicrobial activity	[125]
<i>L. salivarius</i> and <i>L. fermentum</i>	Agar well diffusion assay and spectrophotometric assay	Cell free supernatant	Antimicrobial and antibiofilm activity	[126]
<i>Lactobacillus</i> and <i>Bifidobacterium</i> strains	Agar well diffusion assay	Cell free supernatant	Antimicrobial activity	[127]
<i>L. gasseri</i> YIT 12,321	Agar well diffusion assay	Cell free supernatant	Antimicrobial activity	[128]
<i>L. animalis</i> 30a-2 and <i>W. cibaria</i> C34	Agar well diffusion assay	Cell free supernatant	Antimicrobial activity	[129]
<i>Bacillus subtilis</i> 534	Agar well diffusion assay	Cell free supernatant	Antimicrobial activity	[130]
<i>Lactobacillus acidophilus</i> La-5	Agar well diffusion assay and crystal violet assay	Cell free supernatant	Antimicrobial and antibiofilm activity	[131]
<i>Lactobacillus</i> strains	Agar well diffusion assay and microtitre plate assay	Cell Free Supernatant	Antimicrobial and antibiofilm activity	[132]
<i>Lactobacillus acidophilus</i> NCFM, <i>L. acidophilus</i> CGMCC1.1878	Co-incubation assay	Surface proteins and nisin	Antimicrobial activity	[133]
<i>Lactobacillus plantarum</i> (86), <i>Lactobacillus fermentum</i> (AI2) and <i>Weissella confusa</i> (AI10)	Co-culture	Cell free supernatant	Antimicrobial activity	[134]
<i>Lactobacillus casei</i> Y1 and <i>Lactobacillus plantarum</i>	Tissue culture plate (TCP) method	Cell free supernatant	Antimicrobial and Antibiofilm activity	[72]
<i>Lactobacillus rhamnosus</i> SHA113	In vitro agar well diffusion assay and in vivo studies in mice	Cell free supernatant and live cells	Antimicrobial activity, MRSA exclusion, inhibit the expression of inflammatory factors TNF- α and IL-6, and restore the level of white cells and neutrophils in the blood, up-regulated expression of tight junction proteins ZO-1 and occludin,	[135]
<i>L. brevis</i>	In vivo study in <i>C. elegans</i>	Live cells	MRSA exclusion	[136]
<i>Lactobacillus reuteri</i> ATCC 55,730	In vitro study using keratinocytes	Live cells	Competitive exclusion	[137]
<i>Lactobacillus fermentum</i> strain L23 and <i>L. rhamnosus</i> strain L60	In vitro cell culture assays	Live cells	Ant-adhesion	[103]
<i>Bifidobacterium</i> and <i>Lactobacillus</i>	In vitro assay	Lectin	Antibiofilm	[138]
<i>Lactobacillus acidophilus</i> LA5 and <i>Lactobacillus casei</i> 431	Agar well diffusion assay and spectrophotometric assay	Cell-free supernatant	Antibacterial and antibiofilm	[139]
<i>Lactobacillus rhamnosus</i> GG	keratinocytes	Live cells	Anti-toxic effect	[140]
<i>Lactobacillus casei</i>	In vitro study using bMEC cells	Live cells	Anti-adhesion and Anti-invasion	[141]
<i>Lactobacillus fermentum</i>	Co-incubation assays	Cell free supernatant	Antimicrobial and anti-biofilm	[142]
<i>Lactobacillus plantarum</i>	In vivo	Live cells	Immunomodulatory	[143]
<i>L. rhamnosus</i> GG and <i>L. reuteri</i> DSM 17,938	In vivo	Live cells	Immunomodulatory	[144]
<i>Lactobacillus plantarum</i>	In vitro	LTA	Anti-inflammatory	[145]

Table 1 (continued)

Probiotic strain	Assay	Component	Outcome	References
<i>L. plantarum</i>	In vitro	Live cells and supernatant	Antimicrobial and anti-adhesion effects on extracellular matrix (ECM) proteins such as collagen and mucin	[146]
<i>L. paracasei</i> subsp. <i>paracasei</i>	In vivo	Live cells	Protection against MRSA induced intestinal and colonic mucosae damage	[147]
<i>Weissella paramesenteroides</i> GIR16L4 or <i>Lactobacillus rhamnosus</i> D1	In vitro	Live cells	Anti-toxic effects (Down regulation in the expression of expression of staphylococcal enterotoxins and TSST)	[148]
<i>Lactobacillus fermentum</i> , <i>Lactobacillus acidophilus</i> and <i>Lactobacillus paracasei</i>	In vitro co-culture experiment	Cell free supernatant	Anti-toxic effect (Down regulation in the expression of SEA, SEC and SEE genes)	[149]
<i>L. casei</i> BL23	In vitro cell culture	Sortase A2	Anti-invasion	[150]
<i>L. acidophilus</i> DSM 20,079, <i>L. plantarum</i> DSM 20,079, <i>L. casei</i> ATCC 39,392, <i>L. reuteri</i> ATCC 23,272	In vitro	Live cells	Co-aggregation	[98]
<i>Bacillus subtilis</i> KATMIRA1933 and <i>Bacillus amyloliquefaciens</i> B-1895	In vitro	Live cells	Co-aggregation	[151]
<i>Lactobacillus rhamnosus</i> SHA113	In vivo	Live cells	Up-regulated expression of tight junction proteins ZO-1 and occludin	[135]
<i>Bacillus subtilis</i> (Probiotic strain)	In vivo	Exopolysaccharide	Protection against <i>S. aureus</i> induced inflammation	[152]
<i>Bacillus subtilis</i> (Probiotic strain)	In vivo	Exopolysaccharide	Anti-superantigen induced inflammation	[153]
<i>Lactobacillus kiferi</i>	In vivo	Exopolysaccharide	Immunoprotective action against <i>S. aureus</i>	[154]

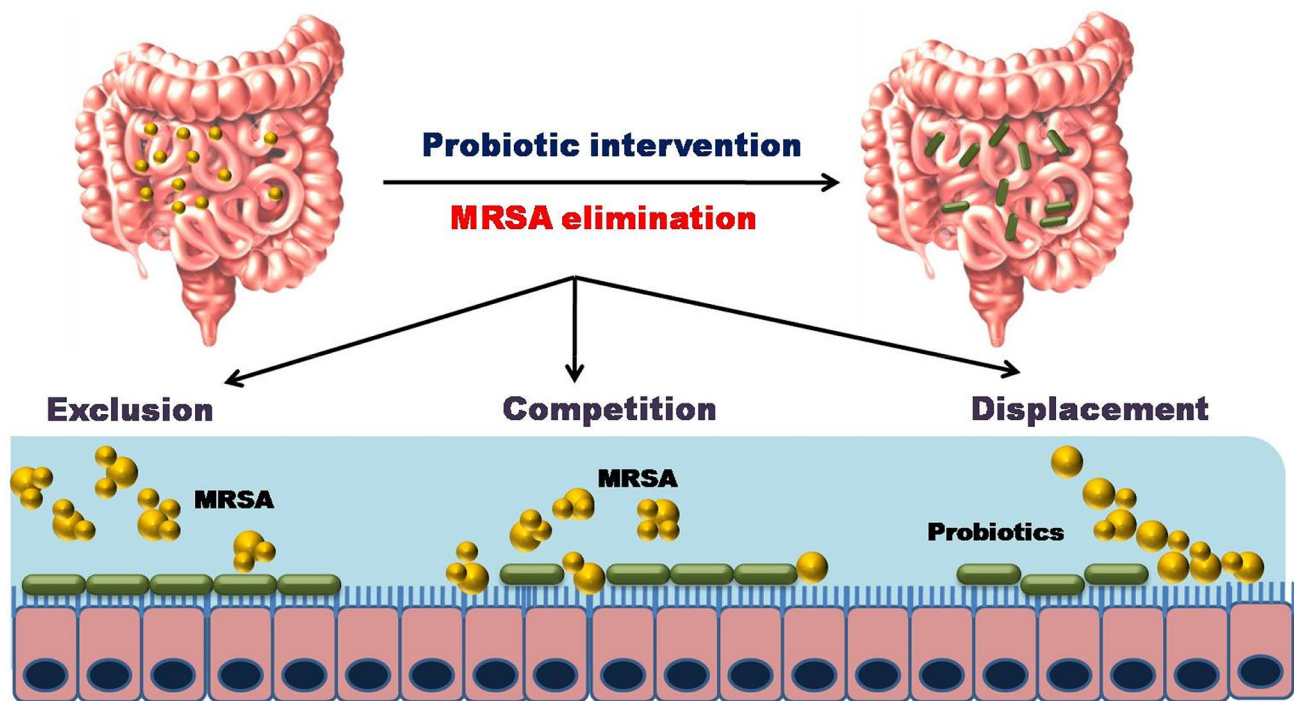


Fig. 2 Pictorial illustration on mode of anti-adhesion knock of probiotics against MRSA

anticolonization efficacy in the competitive assay. Fifteen days of the oral intervention with probiotic *Saccharomyces boulardii* (250 mg) twice daily for a geriatric patient displaying persistent diarrhea and heavy intestinal MRSA growth resulted in successfully eradicated MRSA infection without intestinal recurrence [101]. Moreover, the oral administration of probiotics (*L. rhamnosus* HN001) to intestinal MRSA carrying patients (a randomized study) for 4 weeks revealed the better adherence of probiotics to intestinal milieu with a significant reduction in intestinal colonization of MRSA (50%). More importantly, probiotics were well tolerated in the immunocompromised subjects with multiple comorbidities (diabetes and transplant recipients) [102]. Ortiz et al. [103] highlighted that *Lactobacillus fermentum* L23 is observed to decolonize the *S. aureus* by $19 \pm 13.78\%$, $9 \pm 7.07\%$, and $41 \pm 16.03\%$ in all the three bacterial interferences assays and *L. rhamnosus* L60 exhibited $13.7 \pm 8.38\%$, $6.5 \pm 4\%$, $33.4 \pm 19.05\%$ reduction in the adhesion of *S. aureus* to vaginal epithelial cells. Besides, a human clinical trial conducted by Eggers et al. [104], investigators fed probiotic capsules having *L. rhamnosus* HN001 at 1×10^{10} CFU for 4 weeks to the *S. aureus* carriers. As a result, researchers observed the MRSA shedding in their stool because of pathogen exclusion by ingested probiotics. In another study, the adhesion property of the probiotic strain *L. fermentum* MTCC 8711 and its ability to prevent the adhesion of MRSA to human colon adenocarcinoma cells (Caco-2) was studied. It was reported that a significant

reduction (63%) in the adhesion of MRSA occurred as compared to the MRSA alone in the protective assay. On the other hand, in a competitive assay, a significant reduction (71.1%) in the adhesion of MRSA to Caco-2 was noted. Interestingly, in a displacement assay, approximately 96% reduction in the adhesion of MRSA to the Caco-2 cell line was observed [105].

Probiotics for Probable Enhancement for MRSA Induced Leaky Gut

Intestinal integrity is a major protective mechanism employed to maintain the epithelial barrier between the external lumen environment and the internal milieu. The multiple defense systems of the intestinal barrier consist of the mucous layer, non-specific antimicrobial peptides (alpha defensin, beta-defensins, cathelicidins), secretory IgA, and the epithelial tight junctions, paneth cells, and mast cells [106]. This epithelial structure provides a dynamic and regulated barrier or protection to the flux of luminal contents to the lamina propria. This integrity barrier will be maintained during the healthy gut with the balanced gut microbiota. However, the predomination of enteropathogens in the gut has been found to have proven caliber to disturb the barrier function of the gut by triggering the host inflammatory cytokines [107]. Investigators also reported that the *S. aureus* and their alpha toxins mediated loss of gut barrier

integrity in the human epithelial cell line (Caco-2 cell line). Subsequently, they justified that the disruption in barrier integrity might be due to the potential interplay between toxins and the cytoplasmic calcium ion content. Calcium ions play important roles in cellular functions like vesicular trafficking, regulation of gene expression, cytoskeletal rearrangements, and apoptosis. Any changes in cytosolic free calcium ion concentration might alter the normal cellular responses and regulate the cellular responses [44]. The alpha-toxin perhaps seems to interact with the cytoplasmic membrane and create the pore subsequently replenishing intracellular stored calcium and thereby, alleviate the calcium content, which will eventually affect both tight and adherence junctions factors and will, in turn, abrogate the *S. aureus*-induced septicemia by facilitating the entry of intestinal infectious agents [108]. Once this barrier function is disrupted, bacterial antigens can reach the submucosa and can induce inflammatory responses viz. inflammatory bowel disease (IBD), leaky gut, irritable bowel syndrome (IBS), and colitis leading to increased infiltration of pro-inflammatory cytokines and cytokine-induced changes to the tight junction (TJ). These tight junction proteins viz. occludin, claudins, tricullulin, zonulins zonula occludens (ZO)-1, ZO-2, and ZO-3 maintain the epithelial cells together and control epithelial paracellular permeability. However, any adverse inflammatory effect to TJ might result in simultaneous alteration in epithelial permeability by the disturbance in the expression of different claudins and RNA interference experiments in polarized cell lines coupled with measurements of transepithelial electrical resistance (TER) and dye diffusion across the cells [109].

The consumption of beneficial microorganisms can contribute to strengthening the intestinal barrier function. One such microbial community is “probiotics”, which has been extensively studied for their role in the maintenance of gut barrier function. Although the mechanisms by which probiotics enhance intestinal barrier function are not fully elucidated, few attempts were made to understand their mechanistic pathways. Firstly, balancing and restoration of the intestinal microflora creates protection against pathogens and thus modulates the immune system of the host [110]. Secondly, several studies have confirmed the probiotics induced expression of genes responsible for tight junction signaling [111]. Likewise, *L. rhamnosus* OLL2838 significantly suppressed the barrier permeability created by TNF-alpha both in the in vitro Caco2-cell line model and in vivo mouse model [112]. Further, probiotics also benefit the host in maintaining gut health by promoting mucus secretion by enhancing the expression of *MUC2* and *MUC3* gene as one of the mechanisms to improve barrier function [113]. Although probiotics have shown antiadhesion mechanisms against MRSA, currently there is a limitation in the studies on gut barrier protective efficacy of probiotics in

the MRSA induced intestinal epithelial hyperpermeability model. However, the previous investigation suggests that multi-strains probiotic intervention (*L. acidophilus* NCFM, *Bifidobacterium lactis* 420, *L. acidophilus* La-14, *L. salivarius* Ls-33, and *Propionibacterium jensenii* P63) has shown to regulate the expression of claudin-4, ZO-1, and occludin in the *S. aureus* induced hyperpermeability in human epidermal keratinocytes [114]. In another recent study, the treatment of *Lactobacillus fermentum* NCU3087 and *L. fermentum* NCU3088 attenuated the degradation of occludin, claudin-1, ZO-1, and JAM-1 and suppressed the secretion of IL-6 and tumor necrosis factor- α induced by *S. aureus* treatment in Caco-2 cells treated with *S. aureus* [115]. These findings emphasize the further requirement of in vivo studies to prove the same effects.

Recent Strategies in the Application of Probiotics to Confront MRSA Colonization

Direct oral administration of probiotic fermented products or consumption of viable cells in the form of pills has been frequently practiced and found effective to decolonize MRSA from gut ecosystem [104]. However, the oral administration of probiotic *Lactobacillus rhamnosus* HN001 was not effective to decolonize MRSA from extra-GI sites. In contrast, oral administration of probiotic cells has been reported to reduce intranasal colonization of *S. aureus* [116]. Due to this uncertainty in the findings related to the effectiveness of probiotics to decolonize *S. aureus*, the intranasal administration of probiotics has come into existence to overcome nasal staphylococcal colonization. The development of novel technologies for probiotic nasal spray has made probiotic applications through the intranasal route easier and probiotics preparations from such technologies (spray drying) have been found effective to antagonize *S. aureus* [117]. On the other hand, cryogenic technology has also been used to develop the curative or therapeutic cocktail of multiple mixed strain probiotics and such probiotics have been found helpful to eradicate gut colonization of MRSA in a community-based clinical trial in the hospital settings [118].

Though probiotics are generally encapsulated to reduce the exposure of GI stress and thereby to maintain the viability at the target site, here is a novel strategy that utilizes the same encapsulation principle to overcome antibiotic stress that a probiotic cell undergoes during its co-administration with antibiotics (during adjunct therapies) while combating resistant infections. An attempt was made to co-administer both probiotics and antibiotics to combat MRSA infections without altering the viability of probiotics by biofilm-inspired encapsulation of probiotics to confer temporary antibiotic protection. Probiotic strains viz. *Lactobacillus acidophilus* CL1285, *L. casei* LBC80R,

Table 2 List of clinical trials focused to study the therapeutic and preventive efficacy of probiotics against MRSA (<https://clinicaltrials.gov/>)

Study title	Primary Purpose	Study start date	Study completion date
A Pilot Randomized Trial to Determine the Efficacy of a Probiotic, <i>Lactobacillus rhamnosus</i> for Reducing Colonization by Methicillin-resistant <i>Staphylococcus aureus</i> (MRSA)	Prevention	January 2010	March 2012
Probiotic <i>Bacillus</i> for Decolonization of Methicillin-resistant <i>Staphylococcus aureus</i> (MRSA) in Patients Undergoing Total Joint Arthroplasty: A Randomized, Prospective, Placebo-Controlled Pilot Study	Treatment	January 22, 2021	January 22, 2021
A Single Center, Double-Blind, Randomized, Placebo-Controlled Study Evaluating the Efficacy of Bio-K + Cl-1285® In the Nasal Decolonization of Methicillin Resistant <i>Staphylococcus aureus</i> (MRSA) Carrier Patients	Treatment	January 2010	July 2016

and *L. rhamnosus* CLR2 have been encapsulated within alginate and exposed to the antibiotic tobramycin. The growth and metabolic activity of encapsulated probiotics upon antibiotic exposure were unaffected (four-log survival) and this approach of using tobramycin combined with encapsulated probiotic has ultimately demonstrated its ability to completely eradicate MRSA in co-culture experiments [119].

With the advent of electrospinning technology, the nanoencapsulation of probiotics or their bioactive molecules has gained tremendous interest in recent days [120]. In fact, such probiotic functionalized nanocomposite electrospun scaffold has shown to display better staphylococcal infection control and dermal burn healing in mice models. In addition, intervention with electrospun bioscaffolds has demonstrated re-epithelialization, collagen deposition, and hair follicle formation, inhibition of harmful bacteria, and provided interference benefits [121]. The coaxial utilization of nanofibres for the preparation of nanocomposite scaffolds is a special beauty of this technique since the use of prebiotic (inulin) based nanofibre as a carrier web for the probiotics will improve the viability of probiotics and thereby increases the viability at the target site [122, 123].

Recently, a probiotic-based novel method was developed for orthopedic implants that generally suffer from MRSA colonization during osseointegration and thus contaminate the patients. For successful osseointegration, researchers developed engineered probiotic biofilms by culturing the *Lactobacillus casei* on alkali heat-treated titanium (Ti) substrates and subsequent inactivation of live probiotic cells by ultraviolet irradiation to avoid sepsis. The inactivated *L. casei* biofilm showed 99.98% anti-MRSA activity due to the preformed lactic acid and bacteriocins. Moreover, the polysaccharides in the probiotic biofilm also stimulated the macrophages to secrete

osteogenic cytokines such as oncostatin M with improved osseointegration of Ti implant [124].

Conclusion and Future Perspectives

MRSA is an opportunistic human pathogen that exhibits inherent resistance to penicillin and its derivatives; however, the recent emergence of resistance phenotypes towards vancomycin, carbapenems, and rifampicin antibiotics is highly worrisome. *S. aureus* is inherently a hub of virulome and resistome that assist in escalating a range of life-threatening resistant public health diseases. Albeit MRSA thrives on skin and mucosal milieu, it is not, however, a frequent colonizer of the human gut. Nevertheless, the recent findings on intestinal colonization tactics of MRSA with manifestations of MRSA associated gastroenteritis and colitis (diffuse colitis and mild ileitis) have drawn attention. Therefore, probiotics and their metabolites have been studied for their biotherapeutic ability to combat gut mediated MRSA colonization and subsequent infections. Although probiotics have shown in vitro and in vivo prophylactic actions against MRSA, there are limitations in the clinical trials focusing on the therapeutic efficacy of probiotics against MRSA (Table 2). During the direct application of probiotics to decolonize MRSA, it is also paramount to confirm the absence of horizontal resistance transfer from MRSA to probiotic cells via transconjugant experiments. On the other hand, investigating the safety of probiotics (sepsis or bloodstream infection-causing ability) (Fig. 3) and frequent tests on antibiotic resistance patterns of probiotics are necessary to prevent the probiotics mediated the rampant spread of resistance between the commensal bacteria.

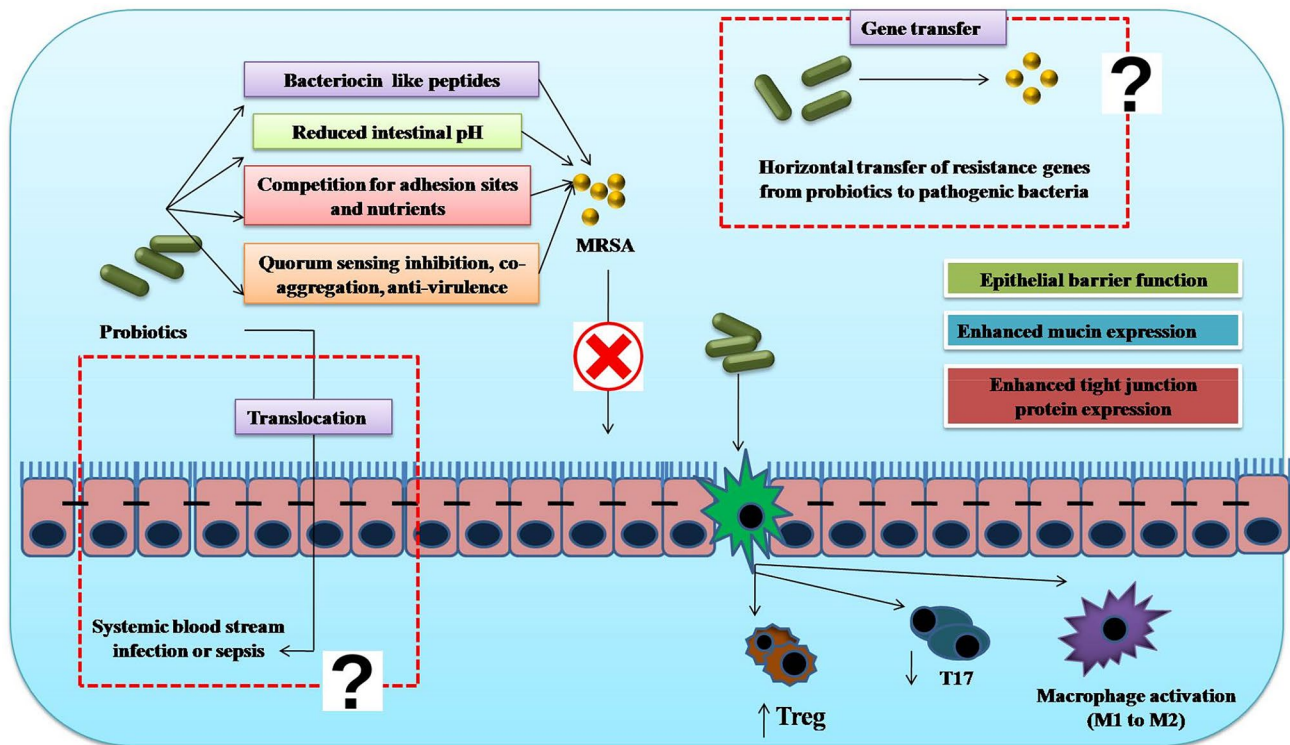


Fig. 3 Illustration on the protective actions of probiotics and possible risks of using live cells of probiotics in combating gut mediated MRSA associated infections

Acknowledgements The authors are highly thankful to the Director, ICAR-National Dairy Research Institute, Karnal.

Authors' Contribution BHN performed the literature review and wrote the manuscript. RHM was a mentor and contributed to the concept of this paper.

Funding Not applicable.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Consent for Publication All the authors are concur with the publication.

References

- Berendonk TU, Manaia CM, Merlin C, Fatta-Kassinos D, Cytryn E, Walsh F, Bürgmann H, Sørum H, Norström M, Pons MN, Kreuzinger N (2015) Tackling antibiotic resistance: the environmental framework. *Nat Rev Microbiol* 13:310
- Ventola CL (2015) The antibiotic resistance crisis: part 1: causes and threats. *Pharm Ther* 40:277
- Pendleton JN, Gorman SP, Gilmore BF (2013) Clinical relevance of the ESKAPE pathogens. *Expert Rev Anti Infect Ther* 11:297–308
- Haaber J, Penadés JR, Ingmer H (2017) Transfer of antibiotic resistance in *Staphylococcus aureus*. *Trends Microbiol* 25:893–905
- Medina E, Pieper DH (2016) Tackling threats and future problems of multidrug-resistant bacteria. *Curr Top Microbiol Immunol* 398:3–33
- Ghosh C, Sarkar P, Issa R, Haldar J (2019) Alternatives to conventional antibiotics in the era of antimicrobial resistance. *Trends Microbiol* 27:323–338
- Henein A (2013) What are the limitations on the wider therapeutic use of phage? *Bacteriophage* 3:e24872
- Mahony J, McAuliffe O, Ross RP, Van Sinderen D (2011) Bacteriophages as biocontrol agents of food pathogens. *Curr Opin Biotechnol* 22:157–163
- <https://cddep.org/research-area/antibiotic-resistance/>
- Stapleton PD, Taylor PW (2002) Methicillin resistance in *Staphylococcus aureus*: mechanisms and modulation. *Sci Progress* 85:57–72
- Nakao A, Ito T, Han X, Lu YJ, Hisata K, Tsujiwaki A, Matsunaga N, Komatsu M, Hiramatsu K, Shimizu T (2014) Intestinal carriage of methicillin-resistant *Staphylococcus aureus* in nasal MRSA carriers hospitalized in the neonatal intensive care unit. *Antimicrob Resist Infect Control* 3:14
- Dong D, Ni Q, Wang C, Zhang L, Li Z, Jiang C, Peng Y (2018) Effects of intestinal colonization by *Clostridium difficile* and *Staphylococcus aureus* on microbiota diversity in healthy individuals in China. *BMC Infect Dis* 18:207

13. Clarke K, Baidoo L (2012) Methicillin-resistant *Staphylococcus aureus* (MRSA) colitis—is there a problem? *Int J Colorect Dis* 27:417–418
14. Karczewski J, Troost FJ, Konings I, Dekker J, Kleerebezem M, Wells BRJM (2010) Regulation of human epithelial tight junction proteins by *Lactobacillus plantarum* *in vivo* and protective effects on the epithelial barrier. *Am J Physiol Gastrointest Liver Physiol* 298:G851–G859
15. Carlet J (2012) The gut is the epicentre of antibiotic resistance. *Antimicrob Resist Infect Control* 1:39
16. Garoy EY, Gebreab YB, Achila OO, Tekeste DG, Kesete R, Ghirmay R, Kiflay R, Tesfu T (2019) Methicillin-Resistant *Staphylococcus aureus* (MRSA): Prevalence and antimicrobial sensitivity pattern among patients—A multicenter study in Asmara, Eritrea. *Can J Infect Dis Med Microbiol* 2019:1–9
17. Valliammai A, Sethupathy S, Ananthi S, Priya A, Selvaraj A, Nivetha V, Aravindraja C, Mahalingam S, Pandian SK (2020) Proteomic profiling unveils citral modulating expression of *IsaA*, *CodY* and *SaeS* to inhibit biofilm and virulence in methicillin-resistant *Staphylococcus aureus*. *Int J Biol Macromol* 158:208–221
18. Infectious Diseases Society of America (IDSA) (2011) Combating antimicrobial resistance: policy recommendations to save lives. *Clin Infect Dis* 52:S397–S428
19. Ito T, Okuma K, Ma XX, Yuzawa H, Hiramatsu K (2003) Insights on antibiotic resistance of *Staphylococcus aureus* from its whole genome: genomic island SCC. *Drug Resist Update* 6:41–52
20. Boundy S, Safo MK, Wang L, Musayev FN, O'Farrell HC, Rife JP, Archer GI (2013) Characterization of the *Staphylococcus aureus* rRNA methyltransferase encoded by *orfX*, the gene containing the *Staphylococcal* chromosome cassette *mec* (*SCCmec*) insertion site. *J Biol Chem* 288:132–140
21. Peacock SJ, Paterson GK (2015) Mechanisms of methicillin resistance in *Staphylococcus aureus*. *Annu Rev Biochem* 84:577–601
22. Shore AC, Coleman DC (2013) *Staphylococcal* cassette chromosome *mec*: recent advances and new insights. *Int J Med Microbiol* 303:350–359
23. Otto M (2012) MRSA virulence and spread. *Cell Microbiol* 14:1513–1521
24. Diep BA, Otto M (2008) The role of virulence determinants in community-associated MRSA pathogenesis. *Trends Microbiol* 16:361–369
25. Salasia SIO, Khusnan Z, Lammler C, Zschock M (2004) Comparative studies on pheno- and genotypic properties of *Staphylococcus aureus* isolated from bovine subclinical mastitis in central Java in Indonesia and Hesse in Germany. *J Vet Sci* 5:103–110
26. Agius P, Kreiswirth B, Naidich S, Bennett K (2007) Typing *Staphylococcus aureus* using the *spa* gene and novel distance measures. *IEEE/ACM Trans Comput Biol Bioinform* 4:693–704
27. Peetermans M, Verhamme P, Vanassche T (2015) Coagulase activity by *Staphylococcus aureus*: a potential target for therapy? *Semin Thromb Hemost* 41:433–444
28. Gill SR, Fouts DE, Archer GL, Mongodin EF, DeBoy RT, Ravel J, Paulsen IT, Kolonay JF, Brinkac L, Beanan M, Dodson RJ (2005) Insights on evolution of virulence and resistance from the complete genome analysis of an early methicillin-resistant *Staphylococcus aureus* strain and a biofilm-producing methicillin-resistant *Staphylococcus epidermidis* strain. *J Bacteriol* 187:2426–2438
29. Al-Ashmary MA, Sallam KI, Abd-Elghany SM, Elhadidy M, Tamura T (2016) Prevalence, molecular characterization, and antimicrobial susceptibility of methicillin-resistant *Staphylococcus aureus* isolated from milk and dairy products. *Foodborne Pathog Dis* 13:156–162
30. Batte JL, Samanta D, Elasmri MO (2016) MsaB activates capsule production at the transcription level in *Staphylococcus aureus*. *Microbiology* 162:575–589
31. Diep BA, Carleton HA, Chang RF, Sensabaugh GF, Perdreau-Remington F (2006) Roles of 34 virulence genes in the evolution of hospital- and community-associated strains of methicillin-resistant *Staphylococcus aureus*. *Journal Infect Dis* 193:1495–1503
32. Cepas V, Lopez Y, Muñoz E, Rolo D, Ardanuy C, Martí S, Xercavins M, Horcajada JP, Bosch J, Soto SM (2019) Relationship between biofilm formation and antimicrobial resistance in Gram-negative bacteria. *Microb Drug Resist* 25:72–79
33. Bridier A, Briandet R, Thomas V, Dubois-Brissonnet F (2011) Resistance of bacterial biofilms to disinfectants: A review. *Biofouling* 27:1017–1032
34. Archer NK, Mazaitis MJ, Costerton JW, Leid JG, Powers ME, Shirtliff ME (2011) *Staphylococcus aureus* biofilms: properties, regulation, and roles in human disease. *Virulence* 2:445–459
35. Kiamco MM, Mohamed A, Reardon PN, Marean-Reardon CL, Aframehr WM, Call DR, Beyenal H, Renslow RS (2018) Structural and metabolic responses of *Staphylococcus aureus* biofilms to hyperosmotic and antibiotic stress. *Biotechnol Bioeng* 115:1594–1603
36. Periasamy S, Joo HS, Duong AC, Bach THL, Tan VY, Chatterjee SS, Cheung GY, Otto M (2012) How *Staphylococcus aureus* biofilms develop their characteristic structure. *Proc Natl Acad Sci* 109:1281–1286
37. Harriott MM, Noverr MC (2009) *Candida albicans* and *Staphylococcus aureus* form polymicrobial biofilms: effects on antimicrobial resistance. *Antimicrob Agents Chemother* 53:3914–3922
38. Dhanawade NB, Kalorey DR, Srinivasan R, Barbudde SB, Kurkure NV (2010) Detection of intercellular adhesion genes and biofilm production in *Staphylococcus aureus* isolated from bovine subclinical mastitis. *Vet Res Commun* 34:81–89
39. Ghafourian S, Mohebi R, Rezaei M, Raftari M, Sekawi Z, Kazemian H, Mohseni A, Karimi S, Sadeghifard N (2012) Comparative analysis of biofilm development among MRSA and MSSA strains. *Roum Arch Microbiol Immunol* 71:175–182
40. Ghasemian A, Peerayeh SN, Bakhshi B, Mirzaei M (2016) Comparison of biofilm formation between methicillin-resistant and methicillin-susceptible isolates of *Staphylococcus aureus*. *Iran Biomed J* 20:175
41. Rooks MG, Garrett WS (2016) Gut microbiota, metabolites and host immunity. *Nat Rev Immunol* 16:341
42. Rowland I, Gibson G, Heinken A, Scott K, Swann J, Thiele I, Tuohy K (2018) Gut microbiota functions: metabolism of nutrients and other food components. *Eur J Nutr* 57:1–24
43. Jandhyala SM, Talukdar R, Subramanyam C, Vuyyuru H, Sasi-kala M, Reddy DN (2015) Role of the normal gut microbiota. *World J Gastroenterol* 21:8787
44. Kwak YK, Vikstrom E, Magnusson KE, Vecsey-Semjen B, Colque-Navarro P, Mollby R (2012) The *Staphylococcus aureus* alpha-toxin perturbs the barrier function in Caco-2 epithelial cell monolayers by altering junctional integrity. *Infect Immun* 80:1670–1680
45. Piewngam P, Zheng Y, Nguyen TH, Dickey SW, Joo HS, Villaruz AE, Gloe KA, Fisher EL, Hunt RL, Li B, Chiou J (2018) Pathogen elimination by probiotic *Bacillus* via signalling interference. *Nature* 562:532–537
46. Basavaprabhu HN, Sonu KS, Prabha R (2020) Mechanistic insights into the action of probiotics against bacterial vaginosis and its mediated preterm birth: An overview. *Microb Pathog* 141:04029

47. Hill C, Guarner F, Reid G, Gibson GR, Merenstein DJ, Pot B, Morelli L, Canani RB, Flint HJ, Salminen S, Calder PC (2014) Expert consensus document: The International Scientific Association for Probiotics and Prebiotics consensus statement on the scope and appropriate use of the term probiotic. *Nat Rev Gastroenterol Hepatol* 11:506
48. O'Toole PW, Marchesi JR, Hill C (2017) Next-generation probiotics: the spectrum from probiotics to live biotherapeutics. *Nat Microbiol* 2:17057
49. Pandey KR, Naik SR, Vakil BV (2015) Probiotics, prebiotics and synbiotics—a review. *J Food Sci Technol* 52(12):7577–7587
50. Nami Y, Haghshenas B, Abdullah N, Barzegari A, Radiah D, Rosli R, Khosroushahi AY (2015) Probiotics or antibiotics: future challenges in medicine. *J Med Microbiol* 64:137–146
51. Nagpal R, Kumar A, Kumar M, Behare PV, Jain S, Yadav H (2012) Probiotics, their health benefits and applications for developing healthier foods: a review. *FEMS Microbiol Lett* 334:1–15
52. Yi L, Dang J, Zhang L, Wu Y, Liu B, Lü X (2016) Purification, characterization and bactericidal mechanism of a broad spectrum bacteriocin with antimicrobial activity against multidrug-resistant strains produced by *Lactobacillus coryniformis* XN8. *Food Control* 67:53–62
53. Satpute SK, Mone NS, Das P, Banat IM, Banpurkar AG (2019) Inhibition of pathogenic bacterial biofilms on PDMS based implants by *L. acidophilus* derived biosurfactant. *BMC Microbiol* 19:1–15
54. Shetty AD, Matthews K, Hegde AR, Akshatha B, Mathias AB, Mutalik S, Vidya SM (2016) Multidrug resistant pathogenic bacterial biofilm inhibition by *Lactobacillus plantarum* exopolysaccharide. *Bioact Carbohydr Diet Fibre* 8:7–14
55. Ogbodo SO, Okeke AC, Ugwuoru CDC, Chukwurah EF (2011) Possible alternatives to reduce antibiotic resistance. *Life Sci Med Res* 2011:1–9
56. Lebeer S, Vanderleyden J, De Keersmaecker SC (2008) Genes and molecules of lactobacilli supporting probiotic action. *Microbiol Mol Biol Rev* 72:728–764
57. Boopathi S, Selvakumar G, Sivakumar N (2017) Quorum quenching potentials of probiotic *Enterococcus durans* LAB38 against methicillin resistant *Staphylococcus aureus*. *Quorum* 10:445–450
58. Chifiriuc MC, Dițu L, Banu O, Bleotu C, Drăcea O, Bucur M, Larion C, Israil AM, Lazăr V (2009) Subinhibitory concentrations of phenyl lactic acid interfere with the expression of virulence factors in *Staphylococcus aureus* and *Pseudomonas aeruginosa* clinical strains. *Roum Arch Microbiol Immunol* 68:27–33
59. Rana S, Bhawal S, Kumari A, Kapila S, Kapila R (2020) pH-dependent inhibition of AHL-mediated quorum sensing by cell-free supernatant of lactic acid bacteria in *Pseudomonas aeruginosa* PAO1. *Microb Pathog* 142:104105
60. Spinler JK, Taweechotipatr M, Rognerud CL, Ou CN, Tumwasorn S, Versalovic J (2008) Human-derived probiotic *Lactobacillus reuteri* demonstrate antimicrobial activities targeting diverse enteric bacterial pathogens. *Anaerobe* 14:166–171
61. Shokryazdan P, Sieo CC, Kalavathy R, Liang JB, Alitheen NB, Faseleh Jahromi M, Ho YW (2014) Probiotic potential of *Lactobacillus* strains with antimicrobial activity against some human pathogenic strains. *BioMed Res Int* 2014:1–16
62. Satpute SK, Kulkarni GR, Banpurkar AG, Banat IM, Mone NS, Patil RH, Cameotra SS (2016) Biosurfactant/s from lactobacilli species: Properties, challenges and potential biomedical applications. *J Basic Microbiol* 56:1140–1158
63. Savadogo A, Ouattara AC, Bassole HI, Traore SA (2006) Bacteriocins and lactic acid bacteria—a minireview. *Afr J Biotechnol* 5:678–683
64. Cotter PD, Ross RP, Hill C (2013) Bacteriocins—a viable alternative to antibiotics? *Nat Rev Microbiol* 11:95
65. Karska-Wysocki B, Bazo M, Smoragiewicz W (2010) Antibacterial activity of *Lactobacillus acidophilus* and *Lactobacillus casei* against methicillin-resistant *Staphylococcus aureus* (MRSA). *Microbiol Res* 165:674–686
66. Saravanakumari P, Mani K (2010) Structural characterization of a novel xylolipid biosurfactant from *Lactococcus lactis* and analysis of antibacterial activity against multi-drug resistant pathogens. *Bioresour Technol* 101:8851–8854
67. Gudiña EJ, Rocha V, Teixeira JA, Rodrigues LR (2010) Antimicrobial and antiadhesive properties of a biosurfactant isolated from *Lactobacillus paracasei* ssp. *paracasei* A20. *Lett Appl Microbiol* 50:419–424
68. Bendjeddou K, Fons M, Strocker P, Sadoun D (2012) Characterization and purification of a bacteriocin from *Lactobacillus paracasei* subsp. *paracasei* BMK2005, an intestinal isolate active against multidrug-resistant pathogens. *World J Microbiol Biotechnol* 28:1543–1552
69. Chen PW, Jheng TT, Shyu CL, Mao FC (2013) Synergistic antibacterial efficacies of the combination of bovine lactoferrin or its hydrolysate with probiotic secretion in curbing the growth of methicillin-resistant *Staphylococcus aureus*. *J Med Microbiol* 62:1845–1851
70. Abbas MM, Mahasneh AM (2014) Isolation of *Lactobacillus* strains with probiotic potential from camels milk. *Afr J Microbiol Res* 8:1645–1655
71. Muralinath E, Parasa LM (2015) *In vitro* investigation on antibacterial activity of selected probiotics against superbug of methicillin resistant *Staphylococcus aureus* (MRSA) isolated from hospital populations. *World J Pharm Pharm Sci* 4:430–445
72. Kumar LM, Saad WZ, Mohamad R, Rahim RA (2017) Influence of biofilm-forming lactic acid bacteria against methicillin-resistant *Staphylococcus aureus* (MRSA S547). *Asian Pac J Trop Biomed* 7:1107–1115
73. Velraeds MM, Van der Mei HC, Reid G, Busscher HJ (1996) Inhibition of initial adhesion of uropathogenic *Enterococcus faecalis* by biosurfactants from *Lactobacillus* isolates. *Appl Environ Microbiol* 62:1958–1963
74. Patowary K, Patowary R, Kalita MC, Deka S (2017) Characterization of biosurfactant produced during degradation of hydrocarbons using crude oil as sole source of carbon. *Front Microbiol* 8:279
75. Sharma D, Saharan BS, Kapil S (2016) Biosurfactants of lactic acid bacteria. Springer, Switzerland
76. Sarikaya H, Aslim B, Yuksekdağ Z (2017) Assessment of anti-biofilm activity and bifidogenic growth stimulator (BGS) effect of lyophilized exopolysaccharides (L-EPSs) from lactobacilli strains. *Int J Food Prop* 20:362–371
77. Walencka E, Rozalska S, Sadowska B, Rozalska B (2008) The influence of *Lactobacillus acidophilus*-derived surfactants on staphylococcal adhesion and biofilm formation. *Folia Microbiol* 53:61
78. Tahmourespour A, Salehi R, Kermanshahi RK, Eslami G (2011) The anti-biofouling effect of *Lactobacillus fermentum*-derived biosurfactant against *Streptococcus mutans*. *Biofouling* 27:385–392
79. Yarwood JM, Schlievert PM (2003) Quorum sensing in Staphylococcus infections. *J Clin Invest* 112:1620–1625
80. Zhang F, Wang B, Liu S, Lin Y, Liu Z, Zhang X, Yu B (2021) *Bacillus subtilis* revives conventional antibiotics against *Staphylococcus aureus* Osteomyelitis. Research Square. <https://doi.org/10.21203/rs.3.rs-136399/v1>
81. Nouaille S, Rault L, Jeanson S, Loubière P, Le Loir Y, Even S (2014) Contribution of *Lactococcus lactis* reducing properties to the downregulation of a major virulence regulator in *Staphylococcus aureus*, the agr system. *Appl Environ Microbiol* 80:7028–7035

82. Li J, Wang W, Xu SX, Magarvey NA, McCormick JK (2011) *Lactobacillus reuteri*-produced cyclic dipeptides quench agr-mediated expression of toxic shock syndrome toxin-1 in staphylococci. *Proc Natl Acad Sci* 108:3360–3365
83. Abbasi M, Dolatabadi S, Ghorbannezhad G, Sharifi F, Rahimi HR (2020) The role of probiotics in inhibition mechanism of methicillin-resistant *Staphylococcus aureus*. *Mcgill J Med* 18:1–11
84. Bholia J, Bhadekar R (2019) In vitro synergistic activity of lactic acid bacteria against multi-drug resistant staphylococci. *BMC Compl Alternative Med* 19:70
85. Ramezani M, Zainodini N, Hakimi H, Rezazadeh Zarandi E, Bagheri V, Bahramabadi R, Zare-Bidaki M (2019) Cell-free culture supernatants of lactobacilli modify the expression of virulence factors genes in *Staphylococcus aureus*. *Jundishapur J Microbiol* 12:e96806
86. Aljasir SF, D'Amico DJ (2020) The effect of protective cultures on *Staphylococcus aureus* growth and enterotoxin production. *Food Microbiol* 91:103541
87. Postollec F, Norde W, De Vries J, Busscher HJ, Van der Mei HC (2006) Interactive forces between co-aggregating and non-co-aggregating oral bacterial pairs. *J Dent Res* 85:231–234
88. Datta A, Stapleton F, Willcox MD (2017) Bacterial coaggregation among the most commonly isolated bacteria from contact lens cases. *Investig Ophthalmol Vis Sci* 58:50–58
89. Van Zyl WF, Deane SM, Dicks LM (2020) Molecular insights into probiotic mechanisms of action employed against intestinal pathogenic bacteria. *Gut Microbes* 12:1831339
90. Hojjati M, Behabehani BA, Falah F (2020) Aggregation, adherence, anti-adhesion and antagonistic activity properties relating to surface charge of probiotic *Lactobacillus brevis* gp104 against *Staphylococcus aureus*. *Microb Pathog* 147:104420
91. Ruiz L, Margolles A, Sánchez B (2013) Bile resistance mechanisms in *Lactobacillus* and *Bifidobacterium*. *Front Microbiol* 4:396
92. Pyclik M, Srutkova D, Schwarzer M, Górski S (2020) Bifidobacteria cell wall-derived exo-polysaccharides, lipoteichoic acids, peptidoglycans, polar lipids and proteins—their chemical structure and biological attributes. *Int J Biol Macromol* 147:333–349
93. Rokana N, Mallappa RH, Batish VK, Grover S (2017) Interaction between putative probiotic *Lactobacillus* strains of Indian gut origin and *Salmonella*: impact on intestinal barrier function. *LWT-Food Sci Technol* 84:851–860
94. Duany RK, Rajput YS, Batish VK, Grover S (2011) Assessing the adhesion of putative indigenous probiotic lactobacilli to human colonic epithelial cells. *Indian J Med Res* 134:664
95. Vemuri R, Shinde T, Shastri MD, Perera AP, Tristram S, Martoni CJ, Gundamaraju R, Ahuja KD, Ball M, Eri R (2018) A human origin strain *Lactobacillus acidophilus* DDS-1 exhibits superior *in vitro* probiotic efficacy in comparison to plant or dairy origin probiotics. *Int J Med Sci* 15:840
96. Neeser JR, Granato D, Rouvet M, Servin A, Teneberg S, Karlsson KA (2000) *Lactobacillus johnsonii* La1 shares carbohydrate-binding specificities with several enteropathogenic bacteria. *Glycobiology* 10:1193–1199
97. Fujiwara S, Hashiba H, Hirota T, Forstner JF (2001) Inhibition of the binding of enterotoxigenic *Escherichia coli* Pb176 to human intestinal epithelial cell line HCT-8 by an extracellular protein fraction containing BIF of *Bifidobacterium longum* SBT2928: suggestive evidence of blocking of the binding receptor ganglioside GM1 on the cell surface. *Int J Food Microbiol* 67:97–106
98. Soleimani NA, Kermanshahi RK, Yakhchali B, Sattari TN (2010) Antagonistic activity of probiotic lactobacilli against *Staphylococcus aureus* isolated from bovine mastitis. *Afr J Microbiol Res* 4:2169–2173
99. Collado MC, Grześkowiak Ł, Salminen S (2007) Probiotic strains and their combination inhibit *in vitro* adhesion of pathogens to pig intestinal mucosa. *Curr Microbiol* 55:260–265
100. Ren D, Li C, Qin Y, Yin R, Li X, Tian M, Du S, Guo H, Liu C, Zhu N, Sun D (2012) Inhibition of *Staphylococcus aureus* adherence to Caco-2 cells by lactobacilli and cell surface properties that influence attachment. *Anaerobe* 18:508–515
101. Sizemore EN, Rivas KM, Valdes J, Caballero J (2012) Enteral vancomycin and probiotic use for methicillin-resistant *Staphylococcus aureus* antibiotic-associated diarrhoea. *Case Rep* 2012:2012006366
102. Warrack S, Panjikar P, Duster M, Safdar N (2014) Tolerability of a probiotic in subjects with a history of methicillin-resistant *Staphylococcus aureus* colonisation. *Benef Microbes* 5:389–395
103. Ortiz L, Ruiz F, Pascual L, Barberis L (2014) Effect of two probiotic strains of *Lactobacillus* on *in vitro* adherence of *Listeria monocytogenes*, *Streptococcus agalactiae*, and *Staphylococcus aureus* to vaginal epithelial cells. *Curr Microbiol* 68:679–684
104. Eggers S, Barker AK, Valentine S, Hess T, Duster M, Safdar N (2018) Effect of *Lactobacillus rhamnosus* HN001 on carriage of *Staphylococcus aureus*: results of the impact of probiotics for reducing infections in veterans (IMPROVE) study. *BMC Infect Dis* 18:129
105. Jayashree S, Karthikeyan R, Nithyalakshmi S, Ranjani J, Gunasekaran P, Rajendhran J (2018) Anti-adhesion property of the potential probiotic strain *Lactobacillus fermentum* 8711 against Methicillin-Resistant *Staphylococcus aureus* (MRSA). *Front Microbiol* 9:411
106. Bermudez-Brito M, Plaza-Díaz J, Muñoz-Quezada S, Gómez-Llorente C, Gil A (2012) Probiotic mechanisms of action. *Ann Nutr Metab* 61:160–174
107. Lamb-Rosteski JM, Kalischuk LD, Inglis GD, Buret AG (2008) Epidermal growth factor inhibits *Campylobacter jejuni*-induced claudin-4 disruption, loss of epithelial barrier function, and *Escherichia coli* translocation. *Infect Immun* 76:3390–3398
108. Berridge MJ, Lipp P, Bootman MD (2000) The calcium entry pas de deux. *Science* 287:1604–1605
109. Krezalek MA, Hyoju S, Zaborin A, Okafor E, Chandrasekar L, Bindokas V, Guyton K, Montgomery CP, Daum RS, Zaborina O, Boyle-Vavra S (2018) Can methicillin-resistant *Staphylococcus aureus* silently travel from the gut to the wound and cause post-operative infection? Modeling the “Trojan Horse Hypothesis.” *Ann Surg* 267:749–758
110. Paton AW, Morona R, Paton JC (2006) Designer probiotics for prevention of enteric infections. *Nat Rev Microbiol* 4:193–200
111. Zyrek AA, Cichon C, Helms S, Enders C, Sonnenborn U, Schmidt MA (2007) Molecular mechanisms underlying the probiotic effects of *Escherichia coli* Nissle 1917 involve ZO-2 and PKC ζ redistribution resulting in tight junction and epithelial barrier repair. *Cell Microbiol* 9:804–816
112. Miyauchi E, Morita H, Tanabe S (2009) *Lactobacillus rhamnosus* alleviates intestinal barrier dysfunction in part by increasing expression of zonula occludens-1 and myosin light-chain kinase *in vivo*. *J Dairy Sci* 92:2400–2408
113. Ohland CL, MacNaughton WK (2010) Probiotic bacteria and intestinal epithelial barrier function. *Am J Physiol Gastrointest Liver Physiol* 298:G807–G819
114. Putaala H, Tiihonen K, Ouwehand AC, Rautonen N (2012) Probiotics modulate tight junction integrity and expression of junctional proteins in cultured normal human epidermal keratinocytes. *Int J Probiotics Prebiotics* 7:81
115. Peng Z, Wei B, Huang T, Liu Z, Guan Q, Xie M, Li H, Xiong T (2020) Screening, safety evaluation, and mechanism of two *Lactobacillus fermentum* strains in reducing the translocation of *Staphylococcus aureus* in the Caco-2 monolayer model. *Front Microbiol* 11:2155

116. Glück U, Gebbers JO (2003) Ingested probiotics reduce nasal colonization with pathogenic bacteria (*Staphylococcus aureus*, *Streptococcus pneumoniae*, and β -hemolytic streptococci). *Am J Clin* 77:517–520
117. Jokicevic K, Kiekens S, Byl E, De Boeck I, Cauwenberghs E, Lebeer S, Kiekens F (2020) Probiotic nasal spray development by spray drying. *Eur J Pharm Biopharm*. <https://doi.org/10.1016/j.ejpb.2020.11.008>
118. Reddy MS, Reddy DRK (2016) Development of multiple mixed strain probiotics for “Probiotic Therapy” under clinical conditions, to prevent or cure the deadly hospital acquired infections due to *Clostridium difficile* (C. diff.) and Methicillin Resistant *Staphylococcus aureus* (MRSA). *Int J Pharm Sci Nanotechnol* 3:3256–3279
119. Li Z, Behrens AM, Ginat N, Tzeng SY, Lu X, Sivan S, Langer R, Jaklenec A (2018) Biofilm-inspired encapsulation of probiotics for the treatment of complex infections. *Adv Mater* 30(51):1803925
120. Wen P, Zong MH, Linhardt RJ, Feng K, Wu H (2017) Electrospinning: a novel nano-encapsulation approach for bioactive compounds. *Trends Food Sci Technol* 70:56–68
121. Khan MA, Hussain Z, Ali S, Qamar Z, Imran M, Hafeez FY (2019) Fabrication of electrospun probiotic functionalized nanocomposite scaffolds for infection control and dermal burn healing in a mice model. *ACS Biomater Sci Eng* 5:6109–6116
122. Wahbi W, Siam R, Kegere J, El-Mehalmey WA, Mamdouh W (2020) Novel inulin electrospun composite nanofibers: prebiotic and antibacterial activities. *ACS Omega* 5:3006–3015
123. Soares J, da Costa MM, de Oliveira HP (2020) Encapsulation and bioavailability of *Lactobacillus* spp. in electrospun fibers. *Curr Biotechnol* 9:15–22
124. Tan L, Fu J, Feng F, Liu X, Cui Z, Li B, Han Y, Zheng Y, Yeung KWK, Li Z, Zhu S (2020) Engineered probiotics biofilm enhances osseointegration via immunoregulation and anti-infection. *Sci Adv*. <https://doi.org/10.1126/sciadv.aba5723>
125. Mertaniasih NM (2015) Antibacterial activity of probiotic mixed culture against MRSA and ESB. *J Chem Pharm Res* 7:1005–1010
126. Kang MS, Lim HS, Oh JS, Lim YJ, Wuertz-Kozak K, Harro JM, Shirliff ME, Achermann Y (2017) Antimicrobial activity of *Lactobacillus salivarius* and *Lactobacillus fermentum* against *Staphylococcus aureus*. *Pathog Dis* 75:1–10
127. Mohamed S, Elmohamady MN, Abdelrahman S, Amer MM, Abdelhamid AG (2020) Antibacterial effects of antibiotics and cell-free preparations of probiotics against *Staphylococcus aureus* and *Staphylococcus epidermidis* associated with conjunctivitis. *Saudi Pharm J* 28:1558–1565
128. Ishikawa T, Imai S, Nakano T, Terai T, Okumura T, Hanada N, Kawahara H (2020) Antibacterial activity of the probiotic candidate *Lactobacillus gasseri* against methicillin-resistant *Staphylococcus aureus*. *APDJ* 20:1–8
129. Lin CF, Lin MY, Lin CN, Chiou MT, Chen JW, Yang KC, Wu MC (2020) Potential probiotic of *Lactobacillus* strains isolated from the intestinal tracts of pigs and feces of dogs with antibacterial activity against multidrug-resistant pathogenic bacteria. *Arch Microbiol* 202:1849–1860
130. Efremenkova O, Gabrielyan N, Malanicheva I, Demiankova M, Efimenko T, Rogozhin E, Sharapchenko S, Krupenyo T, Davydov D, Kornilov M (2019) Antimicrobial properties of the probiotic strain *Bacillus subtilis* 534. *Int Arch Med Microbiol* 2:119
131. Salahuddin K, Salma M (2015) *Lactobacillus acidophilus* LA-5 inhibits growth and biofilm formation of methicillin resistant *Staphylococcus aureus* isolated from wounds of hospitalized patients. *Research and Reviews in Biosciences* 10:300–304
132. Khalil MA, Allam NG, El-Mahallawy DS (2019) Efficacy of *Lactobacillus* spp. in inhibiting the biofilm skin infections induced by *Staphylococcus aureus* pathogen. *Biomed Res*. <https://doi.org/10.35841/biomedicalresearch.30-17-390>
133. Wang H, Niu Y, Pan J, Li Q, Lu R (2020) Antibacterial effects of *Lactobacillus acidophilus* surface-layer protein in combination with nisin against *Staphylococcus aureus*. *LWT-Food Sci Technol* 124:109208
134. Shah N, Patel A, Ambalam P, Holst O, Ljungh A, Prajapati J (2016) Determination of an antimicrobial activity of *Weissella confusa*, *Lactobacillus fermentum*, and *Lactobacillus plantarum* against clinical pathogenic strains of *Escherichia coli* and *Staphylococcus aureus* in co-culture. *Ann Microbiol* 66:1137–1143
135. Liu G, Pang B, Li N, Jin H, Li J, Wu W, Ai C, Jiang C, Shi J (2020) Therapeutic effect of *Lactobacillus rhamnosus* SHA113 on intestinal infection by multi-drug-resistant *Staphylococcus aureus* and its underlying mechanisms. *Food Funct* 11:6226–6239
136. Hesselager MO, Harders RH, Mørch MGM, Møller KV, Sørensen RH, Olsen A (2018) *C. elegans* reveals probiotic secrets: *L. brevis* increases lifespan via the p38 MAP kinase PMK-1. International Scientific Conference on Probiotics, Prebiotics, Gut Microbiota and Health.
137. Prince T, McBain AJ, O'Neill CA (2012) *Lactobacillus reuteri* protects epidermal keratinocytes from *Staphylococcus aureus*-induced cell death by competitive exclusion. *Appl Environ Microbiol* 78:5119–5126
138. Lakhtin M, Alyoshkin V, Lakhtin V, Afanasyev S, Pozhalostina L, Pospelova V (2010) Probiotic lactobacillus and bifidobacterial lectins against *Candida albicans* and *Staphylococcus aureus* clinical strains: new class of the pathogen biofilm destructors. *Probiotics Antimicrob Proteins* 2:186–196
139. Koohestani M, Moradi M, Tajik H, Badali A (2018) Effects of cell-free supernatant of *Lactobacillus acidophilus* LA5 and *Lactobacillus casei* 431 against planktonic form and biofilm of *Staphylococcus aureus*. *Vet Res Forum* 9:301
140. Mohammadsaeed W, McBain AJ, Cruickshank SM, O'Neill CA (2014) *Lactobacillus rhamnosus* GG inhibits the toxic effects of *Staphylococcus aureus* on epidermal keratinocytes. *Appl Environ Microbiol* 80:5773–5781
141. Bouchard DS, Rault L, Berkova N, Le Loir Y, Even S (2013) Inhibition of *Staphylococcus aureus* invasion into bovine mammary epithelial cells by contact with live *Lactobacillus casei*. *Appl Environ Microbiol* 79:877–885
142. Varma P, Nisha N, Dinesh KR, Kumar AV, Biswas R (2011) Anti-infective properties of *Lactobacillus fermentum* against *Staphylococcus aureus* and *Pseudomonas aeruginosa*. *J Mol Microbiol Biotechnol* 20:137–143
143. Ren D, Gong S, Shu J, Zhu J, Rong F, Zhang Z, Wang D, Gao L, Qu T, Liu H, Chen P (2017) Mixed *Lactobacillus plantarum* strains inhibit *Staphylococcus aureus* induced inflammation and ameliorate intestinal microflora in mice. *BioMed Res Int* 2017:1–7
144. Johansson MA, Björkander S, Mata Forsberg M, Qazi KR, Salvany Celades M, Bittmann J, Eberl M, Sverremark-Ekström E (2016) Probiotic lactobacilli modulate *Staphylococcus aureus*-induced activation of conventional and unconventional T cells and NK cells. *Front Immunol* 7:273
145. Kim HG, Lee SY, Kim NR, Ko MY, Lee JM, Yi TH, Chung SK, Chung DK (2008) Inhibitory effects of *Lactobacillus plantarum* lipoteichoic acid (LTA) on *Staphylococcus aureus* LTA-induced tumor necrosis factor- α production. *J Microbiol Biotechnol* 18:1191–1196
146. Mukherjee S, Ramesh A (2015) Bacteriocin-producing strains of *Lactobacillus plantarum* inhibit adhesion of *Staphylococcus aureus* to extracellular matrix: quantitative insight and implications in antibacterial therapy. *J Med Microbiol* 64:1514–1526

147. Bendali F, Madi N, Sadoun D (2011) Beneficial effects of a strain of *Lactobacillus paracasei* subsp. *paracasei* in *Staphylococcus aureus*-induced intestinal and colonic injury. *Int J Infect Dis* 15:787–e794
148. Silva GO, Castro RD, Oliveira LG, Sant'Anna FM, Barbosa CD, Sandes SHC, Silva RS, Resende MFS, Lana AMQ, Nunes AC, Cerqueira MMOP (2020) Viability of *Staphylococcus aureus* and expression of its toxins (SEC and TSST-1) in cheeses using *Lactobacillus rhamnosus* D1 or *Weissella paramesenteroides* GIR16L4 or both as starter cultures. *J Dairy Sci* 103:4100–4108
149. Misaghi A, Parsaeimehr M, Akhondzadeh A, Zahraee Salehi T, Gandomi H, Azizkhani M (2017) The inhibitory effects of *Lactobacillus fermentum*, *Lactobacillus acidophilus* and *Lactobacillus paracasei* isolated from yoghurt on the growth and enterotoxin A gene expression of *S. aureus*. *Iran J Vet Med* 11:191–201
150. Souza RF, Jardim J, Cauty C, Rault L, Bouchard DS, Bermúdez-Humarán LG, Langella P, Monedero V, Seyffert N, Azevedo V, Le Loir Y (2017) Contribution of sortase SrtA2 to *Lactobacillus casei* BL23 inhibition of *Staphylococcus aureus* internalization into bovine mammary epithelial cells. *PLoS ONE* 12:0174060
151. Algburi A, Al-Hasani HM, Ismael TK, Abdelhameed A, Weeks R, Ermakov AM, Chikindas ML (2020) Antimicrobial activity of *Bacillus subtilis* KATMIRA1933 and *Bacillus amyloliquefaciens* B-1895 against *Staphylococcus aureus* biofilms isolated from wound infection. *Probiotics Antimicrob Proteins*. <https://doi.org/10.1007/s12602-020-09673-4>
152. Paik W, Alonzo F, Knight KL (2019) Probiotic exopolysaccharide protects against systemic *Staphylococcus aureus* infection, inducing dual-functioning macrophages that restrict bacterial growth and limit inflammation. *Infect Immun* 87:e00791–e818
153. Paik W, Alonzo F, Knight KL (2020) Suppression of *Staphylococcus aureus* superantigen-independent interferon gamma response by a probiotic polysaccharide. *Infect Immun*. <https://doi.org/10.1128/IAI.00661-19>
154. Xiu L, Sheng S, Hu Z, Liu Y, Li J, Zhang H, Liang Y, Du R, Wang X (2020) Exopolysaccharides from *Lactobacillus kiferi* as adjuvant enhanced the immuno-protective against *Staphylococcus aureus* infection. *Int J Biol Macromol* 161:10–23

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